

Weighted Farey Laplacians

Bounded Operators, Finite Approximation, a Universal Projection Identity for the Mertens Function, and Near-Sharp Gap Bounds

Author: Micah Blumberg

Self Aware Networks Research Institute

<https://selfawarenetworks.com>

July 2026

Abstract

An unweighted Laplacian on the full Farey graph fails before any spectral or arithmetic interpretation can begin: every vertex has infinite degree, so the natural quadratic form is infinite on every nonzero finitely supported vector. This paper solves that existence problem with denominator-damped edge weights. For weights $w_{uv} = (q_u - q_v)^{-\alpha}$, the threshold is exact: the form fails for $\alpha \leq 1$, whereas for $\alpha > 1$ it defines a bounded positive self-adjoint operator. Finite Farey-level truncations converge strongly, with an explicit per-vertex tail estimate, and magnetic phases enter finite spectral traces only through cycle holonomies.

At finite level, the constant zero mode and the position unitary recover the classical Mertens exponential sum. The resulting commutator formula is derived in its strongest correct form: for every finite-dimensional self-adjoint L with $L\phi_0 = 0$, the vector $L^\perp + [U, L]\phi_0$ is exactly the projection $-(I - P_0)U\phi_0$. The associated Pythagoras identity therefore depends on the kernel projection, not on the positive eigenvalues of the weighted Laplacian. Applied to the Farey graph, its deficit is $[(M(N)+1)/|F_N|]^2$, giving an exact restatement of the classical Mertens criterion for the Riemann Hypothesis but no new proof mechanism by itself.

The paper also proves a uniform trace-class bound for the canonical commutator, an $O(N^{-4})$ upper bound for the first positive eigenvalue, and - by reproducing the smaller-parent canonical-path argument - a lower bound of order $1/(N^4 \log N)$ with an unspecified absolute constant. A global-trace no-go theorem shows that fixed spectral traces in the stated class cannot carry the required Mertens-scale signal. The surviving contribution is consequently precise and bounded: a well-defined arithmetic graph-operator family, controlled finite approximation, near-sharp gap scale, and a transparent account of which operator observables do and do not retain arithmetic information.

1. The Problem Before the Operator

A graph Laplacian is often written down as though its existence were automatic. On a finite graph that instinct is usually harmless. On the full Farey graph it is wrong. Every reduced rational has infinitely many unimodular neighbors. With unit edge weights, a point mass at one vertex pays a positive amount of energy on infinitely many incident edges. The proposed quadratic form is therefore infinite before one can ask about eigenvalues, spectral gaps, magnetic phases, or the Riemann Hypothesis.

The first problem is thus ordinary and concrete:

What decay of the edge weights is sufficient - and necessary - to turn the natural Farey incidence form into a legitimate operator?

Once that is solved, three further questions become meaningful:

1. Do finite denominator cutoffs approximate the infinite operator?
2. Which phase information survives gauge transformations?
3. Does an observable of the finite operator encode a classical arithmetic quantity?

The answers are: denominator decay with exponent greater than one is necessary and sufficient for this form construction; finite-edge truncations converge strongly; magnetic phase dependence is organized by cycle holonomy; and the constant-mode expectation of a position unitary gives the classical Mertens sum.

The last result must be interpreted carefully. The original presentation made the commutator identity look dependent on the weighted Laplacian's positive spectrum. Exact operator algebra shows that dependence cancels. The identity is a universal kernel-projection formula. That correction does not erase the valid operator theory. It identifies its proper mathematical boundary.

1.1 Source lineage

This operator family was originally developed inside the HSIT and SIT / Riemann research program as an attempt to turn a diagonal-plus-adjacency sketch into a value-defined mathematical object. The Farey edge-phase incidence construction was the concrete candidate selected in that program. The theorems below concern the explicit operator whether or not that motivating interpretation succeeds. This distinction preserves the research genealogy without treating SIT motivation as a derivation of the weights, the graph, or a Hilbert-Polya operator.

2. Farey Geometry and the Weighted Incidence Form

2.1 Vertices, edges, and level cutoffs

Let F_N be the reduced fractions p/q in $[0, 1]$ with $q \leq N$, including both $0/1$ and $1/1$. Let F be their union. Its cardinality is

$$m(N) = |F_N| = 1 + \sum_{q \leq N} \varphi(q) = \frac{3}{\pi^2} N^2 + O(N \log N). \quad (1)$$

Some earlier program notes used the half-open set obtained by deleting $0/1$; that convention yields $M(N)$ rather than $M(N)+1$ in the exponential sum. This paper fixes the two-endpoint convention above throughout. The difference is bookkeeping, not a competing arithmetic claim.

Two distinct reduced fractions p/q and r/s are Farey neighbors when

$$|ps - rq| = 1. \quad (2)$$

Write G_N for the induced graph on F_N . Every edge is oriented from the smaller fraction u to the larger fraction v . The orientation fixes signs; it does not change the underlying undirected graph.

2.2 The operation before the terminology

Suppose a complex amplitude is assigned to every rational vertex. An edge compares the amplitudes at its two endpoints after applying opposite phase shifts. The comparison is multiplied by the square root of an edge weight. Summing the squared magnitudes of all such comparisons gives the energy of the configuration.

That operation is the weighted magnetic incidence form. For an oriented edge $e=(u \rightarrow v)$, define

$$(A\psi)_e = \sqrt{w_e} \left(e^{i\theta_e} \psi(u) - e^{-i\theta_e} \psi(v) \right), \quad q_L[\psi] = \|A\psi\|^2. \quad (3)$$

Initially ψ is finitely supported. When the form is bounded, its representing operator is the weighted magnetic Laplacian $L = A^* A$.

The denominator family studied here is

$$w_e = (q_u q_v)^{-\alpha}, \quad \alpha \geq 0. \quad (4)$$

The canonical case is $\alpha=2$. For Farey neighbors, $|u-v|=1/(q_u q_v)$, so the canonical weight is the square of their separation on the real line. It is also the natural product scale associated with tangent Ford circles. These geometric observations motivate the choice; they do not derive a physical Hamiltonian or establish Super Information Theory.

Expanding the form gives diagonal entries equal to the weighted degree

$$W_\alpha(x) = \sum_{e \ni x} w_e, \quad (5)$$

and, for an oriented edge, off-diagonal entries $L_{uv} = -w_e \exp(-2i\theta_e)$ and $L_{vu} = \text{conj}(L_{uv})$.

3. The Exact Existence Threshold

3.1 Infinite degree

Every vertex of the full Farey graph has infinite degree. For a reduced p/q , Bézout's identity supplies integers r_0, s_0 with $p s_0 - r_0 q = 1$. The fractions

$$\frac{r_0 + tp}{s_0 + tq}, \quad t = 0, 1, 2, \dots \quad (6)$$

are reduced Farey neighbors for all sufficiently large t , and their denominators are distinct. Endpoint representatives can be handled directly: $0/1$ is adjacent to every $1/s$, and $1/1$ to every $(s-1)/s$.

3.2 Denominator families

Fix p/q . Solving $p s - r q = \pm 1$ shows that, for each sign, admissible neighbor denominators occupy one residue class modulo q ; for a given denominator s , the numerator is unique. There are therefore at most two neighbors for each s . Consequently,

$$W_\alpha(p/q) \leq 2\zeta(\alpha)q^{-\alpha}, \quad \alpha > 1. \quad (7)$$

For $\alpha \leq 1$, each nonempty residue family gives a divergent harmonic-type sum. This yields the threshold theorem.

Theorem 1 (failure for slow decay). If $\alpha \leq 1$, the form q_L is infinite on every nonzero finitely supported vector. Its finiteness domain is $\{0\}$, so it does not define a densely defined quadratic form on $\ell^2(F)$.

Proof. Choose a vertex v where a finitely supported ψ is nonzero. All but finitely many neighbors of v lie outside the support. Each corresponding edge contributes $w_e |\psi(v)|^2$, and the incident weight sum diverges. square

Theorem 2 (bounded positive operator). If $\alpha > 1$, the form extends to all of $\ell^2(F)$ and defines a bounded positive self-adjoint operator with

$$\|L\| \leq 2 \sup_v W_\alpha(v) \leq 4\zeta(\alpha). \quad (8)$$

At $\alpha=2$, $\|L\| \leq 2\pi^{2/3}$.

Proof. The elementary inequality $|a-b|^2 \leq 2(|a|^2 + |b|^2)$ gives

$$q_L[\psi] \leq 2 \sum_v W_\alpha(v) |\psi(v)|^2 \leq 2 \sup_v W_\alpha(v) \|\psi\|^2. \quad (9)$$

The bounded positive form has a unique bounded positive self-adjoint representative. If V is any real self-adjoint diagonal multiplication operator, then $V+L$ is self-adjoint on $D(V)$ because L is bounded. square

This is an operator-existence result. Calling $V+L$ a Hamiltonian is mathematically permissible, but no physical identification follows without a separate derivation connecting V , the graph, and the weights to measured physics.

4. Finite Approximation and Magnetic Structure

4.1 Tail control

Let L_N contain only edges whose endpoints lie in F_N , extended by zero on the orthogonal complement. For $v=p/q$ in F_N , let $T_N(v)$ be the total weight of its omitted incident edges. The other endpoint of every omitted edge has denominator $s > N$; hence

$$T_N(p/q) \leq 2q^{-\alpha} \sum_{s>N} s^{-\alpha} \leq \frac{2}{(\alpha-1)q^\alpha N^{\alpha-1}}. \quad (10)$$

In the canonical case, $T_N(p/q) \leq 2/(q^2 - N)$.

Theorem 3 (strong finite-edge approximation). For every $\alpha > 1$, $L_N \psi \rightarrow L \psi$ for every ψ in $\ell^2(F)$. If V is a common self-adjoint diagonal operator, then $V+L_N$ converges to $V+L$ in the strong resolvent sense.

Proof. Treat A_N as the incidence operator with zero rows on omitted edges. For finitely supported ψ ,

$$\|(A - A_N)\psi\|^2 \leq 2 \max_{v \in \text{supp } \psi} T_N(v) \|\psi\|^2 \rightarrow 0. \quad (11)$$

The operators A and A_N are uniformly bounded. Expanding $L - L_N = A^* (A - A_N) + (A - A_N)^* A_N$ therefore gives convergence on finitely supported vectors. Uniform boundedness and density extend it to all of $\ell^2(F)$. The resolvent identity then gives strong resolvent convergence after adding the common V , vector by vector. square

The estimate is a denominator-cutoff estimate. It is not a continuum limit and does not assign physical units.

4.2 Gauge covariance and holonomy

The phase attached to one edge is not itself invariant. Let χ_x be a phase at every vertex and let $D = \text{diag}(\exp(-i\chi_x))$. Under

$$\theta_e \mapsto \theta_e + \frac{\chi_u - \chi_v}{2}, \quad (12)$$

the Laplacian changes by $L \rightarrow D^{-1} L D$. Thus its spectrum is unchanged.

At finite level,

$$\text{tr}(L_N^k) = \sum_{v_0, \dots, v_k = v_0} \prod_{j=0}^{k-1} (L_N)_{v_j v_{j+1}}. \quad (13)$$

Every term is a closed walk. Vertex phase changes telescope along that walk, so the surviving phase is the signed sum around its cycles: the holonomy. This is the correct gauge-invariant content of the magnetic phase field.

5. The Arithmetic Layer

Set all edge phases to zero. Because consecutive Farey fractions are neighbors, G_N is connected. The kernel of L_N is therefore the constant line. Define

$$\phi_0 = m^{-1/2} \mathbf{1}, \quad U(\rho) = e^{2\pi i \rho}. \quad (14)$$

The expectation of U in the constant mode is an average over F_N . The Ramanujan-sum identity $c_q(1) = \mu(q)$ gives

$$\sum_{\rho \in F_N} e^{2\pi i \rho} = 1 + \sum_{q \leq N} c_q(1) = 1 + M(N), \quad (15)$$

where $M(N) = \sum_{q \leq N} \mu(q)$ is the Mertens function. Hence

$$\langle \phi_0, U \phi_0 \rangle = \frac{M(N) + 1}{m}. \quad (16)$$

Equation [16] is classical arithmetic. It does not depend on the positive spectrum, edge weights, or spectral gap. The weighted graph supplies a carrier with the correct constant kernel, but it does not create the exponential sum.

6. The Universal Projection Identity

The relevant operation can be understood before naming the formula. Apply the position unitary to the constant state. That creates a vector with one component parallel to the kernel and another orthogonal to it. The commutator with any self-adjoint operator that annihilates the constant state produces the image of the orthogonal component under that operator. The pseudoinverse then cancels the operator on its range. What remains is only the orthogonal projection.

Theorem 4 (universal kernel-projection identity). Let L be a finite-dimensional self-adjoint operator, let P_0 project onto $\ker L$, let L^+ be the Moore-Penrose pseudoinverse, and let ϕ_0 be a unit vector in $\ker L$. For every linear operator U ,

$$L^+[U, L]\phi_0 = -(I - P_0)U\phi_0. \quad (17)$$

Proof. Since $L\phi_0=0$, $[U, L]\phi_0=-L U\phi_0$. Self-adjointness gives $L^++L=I-P_0$. Substitution proves [17]. $\square$

If U is unitary and the kernel is the one-dimensional span of ϕ_0 , the parallel and orthogonal components of $U\phi_0$ satisfy

$$|\langle \phi_0, U\phi_0 \rangle|^2 + \|L^+[U, L]\phi_0\|^2 = 1. \quad (18)$$

This is the commutator-Pythagoras identity in its corrected form. Changing every positive eigenvalue while preserving the kernel leaves the right side of [17] unchanged. The identity is therefore not a new Laplacian-specific spectral mechanism.

For the Farey position unitary, define

$$\delta(N) = 1 - \|L_N^+[U, L_N]\phi_0\|^2. \quad (19)$$

Equations [16] and [18] give the exact equality

$$\delta(N) = \left(\frac{M(N) + 1}{m(N)} \right)^2. \quad (20)$$

Using [1] and the classical Mertens criterion,

$$\text{RH} \iff \delta(N) = O(N^{-3+\varepsilon}) \text{ for every } \varepsilon > 0. \quad (21)$$

Equation [21] is a correct equivalence. It is also transparent about the remaining difficulty: proving the rate is exactly as hard as obtaining the corresponding Mertens bound. The identity does not smuggle positive-spectrum information into that estimate.

The unconditional oscillation $M(N)=O(N^{1/2})$ and $O(N^{1/2})$ also implies that $\delta(N)=O(N^{-3})$ along subsequences. Accordingly, the RH-grade exponent in [21] cannot be replaced by a uniformly faster power. More generally, a bound $\delta(N)=O(N^{-\delta_0})$ would give $M(N)=O(N^{(2-\delta_0)/2})$; any fixed $\delta_0>2$ would already be a power-saving Mertens estimate beyond currently known unconditional bounds.

7. A Uniform Trace-Class Commutator

The universal cancellation in Section 6 does not make every operator estimate vacuous. For the canonical weights, the full commutator is trace class.

For an edge $e=\{u, v\}$, let $\delta_e=\exp(2\pi i u)-\exp(2\pi i v)$. Its rank-two contribution to $[U, L]$ has two singular values equal to $w_e|\delta_e|$, and therefore trace norm $2w_e|\delta_e|$. Farey adjacency gives $|u-v|=1/(q_u - q_v)$, so

$$\|T_e\|_{S_1} \leq 4\pi(q_u q_v)^{-3}. \quad (22)$$

Summing by incident denominator families and then over all vertices yields

$$\|[U, L]\|_{S_1} \leq 4\pi[\zeta(2) + \zeta(3)]. \quad (23)$$

Indeed, the two-neighbors-per-denominator bound gives an incident sum at most $2 \zeta(3) q_v^{-3}$; double counting edges contributes one half; and

$$\sum_{v \in F} q_v^{-3} = 1 + \sum_{q \geq 1} \varphi(q) q^{-3} = 1 + \frac{\zeta(2)}{\zeta(3)}. \quad (24)$$

The series of edge operators converges absolutely in trace norm, so [23] holds both for every finite level and for the infinite canonical operator. This estimate controls the commutator as an operator ideal element. It does not restore positive-spectrum dependence to [17].

8. Near-Sharp Spectral-Gap Scale

Let $\lambda_2(L_N)$ be the smallest positive eigenvalue of the zero-phase canonical Laplacian.

8.1 Upper bound from a deep convergent

For any vertex v , the vector δ_{v-m-1} is orthogonal to the constant mode. Its Rayleigh quotient is $W(v)/(1-1/m)$. Choose the deepest Fibonacci convergent F_k/F_{k+1} with denominator at most N . Its denominator is comparable to N ; its level- N neighbors have denominators comparable to N ; and the number of those neighbors is bounded. Thus its weighted degree is $O(N^{-4})$.

Theorem 5 (gap upper bound).

$$\lambda_2(L_N) \leq CN^{-4}. \quad (25)$$

for an absolute constant C and all sufficiently large N .

8.2 Lower bound from smaller-parent routing

Every non-endpoint Farey fraction is the mediant of two parents. If its denominator is $q=q'+q''$, choose as its routing parent the parent with smaller denominator. That denominator is at most $q/2$. Iterating this rule gives a rooted forest joined by the edge between $0/1$ and $1/1$; every route reaches the root region in $O(\log N)$ steps.

For a vertex v of denominator Q , let $T(v)$ contain the vertices whose smaller-parent route passes through v . A child for which v is the smaller parent has denominator at least $2Q$. At depth j , denominators are therefore at least $2^j Q$. Because adjacent Farey fractions with denominators a, b are separated by $1/(ab)$, all descendants lie in an interval of length $O(Q^{-2})$. Consecutive fractions in F_N are separated by at least N^{-2} ; hence

$$|T(v)| \leq 1 + C_0 \frac{N^2}{Q^2}. \quad (26)$$

Use the path from x to the root region and then back down to y . The path length is at most $C_1 \log N$. For a tree edge between denominators Q and s , where $s \leq Q/2$, at most $2|T(v)|m$ ordered pairs use the edge. Its inverse conductance is $(Qs)^{-2}$. The canonical-path congestion is therefore bounded by

$$(Qs)^{-2} [2|T(v)|m] [C_1 \log N] \leq C_2 m N^4 \log N. \quad (27)$$

The root edge has at most $O(m^2 \log N)$ congestion, which is smaller than the bound in [27] because $m = \Theta(N^2)$.

For the unnormalized weighted Laplacian, the canonical-path Poincaré inequality is

$$\lambda_2(L_N) \geq \frac{2m}{\max_e \left\{ w_e^{-1} \sum_{x,y: e \in \gamma_{xy}} |\gamma_{xy}| \right\}}. \quad (28)$$

Combining [27] and [28] proves:

Theorem 6 (near-sharp gap lower bound). There is an unspecified absolute constant $c > 0$ such that

$$\lambda_2(L_N) \geq \frac{c}{N^4 \log N}. \quad (29)$$

Together,

$$\frac{c}{N^4 \log N} \leq \lambda_2(L_N) \leq \frac{C}{N^4}. \quad (30)$$

The logarithmic factor is the remaining gap in this method. No explicit value of c is derived here. The lower-bound proof is a finite weighted-graph theorem; it does not prove RH, HSIT, or any zeta-zero spectral identification.

9. Which Spectral Summaries Cannot Carry the Signal

The finite trace of the weighted Laplacian is the sum of its weighted degrees. For the general exponent family,

$$\text{tr}(L_N) \leq 2\zeta(\alpha) \left[1 + \sum_{q \leq N} \varphi(q) q^{-\alpha} \right]. \quad (31)$$

Thus it is $O(1)$ for $\alpha > 2$, $O(\log N)$ for $\alpha = 2$, and $O(N^{-(2-\alpha)})$ for $1 < \alpha < 2$.

Let g be fixed and continuous on the common spectral interval, with $g(0) = 0$ and $g(x) = O(x)$ near zero. Compactness makes $|g(x)| \leq C_g x$ throughout the interval. Spectral calculus gives

$$|\text{tr } g(L_N)| \leq C_g \text{tr}(L_N). \quad (32)$$

In the canonical case every such trace grows at most logarithmically. It therefore cannot equal a target known to have $\Omega(N^{1/2})$ subsequences, such as the classical Farey discrepancy linked to the Mertens function.

This is a scoped no-go result. It does not exclude:

- the position-dependent expectation in [16];
- off-diagonal matrix elements;
- observables that depend explicitly on N ;
- unbounded functions of the operator;
- phase-dependent holonomy observables; or
- arithmetic information inserted through a separate diagonal or positional operator.

The distinction matters. The arithmetic signal in [20] survives precisely because U remembers where each rational lies; a fixed trace of L_N alone forgets that positional information.

10. Relation to Earlier RH Reformulations

The construction belongs to a large family of exact reformulations rather than to a completed proof program. Nyman-Burling and Báez-Duarte express RH through approximation in a Hilbert space. Li and Bombieri-Lagarias use positivity criteria. Connes uses a trace-formula framework. Mayer, Lewis-Zagier, Bonanno, and Isola connect Gauss or Farey dynamics to transfer operators and modular spectral data. Redheffer-type matrices encode the Mertens function in finite divisibility matrices.

The present carrier differs from those constructions: it is an ℓ^2 space over rational vertices with a self-adjoint weighted graph Laplacian and a diagonal position unitary. Its useful distinctions are:

1. the infinite operator has an exact, proved existence threshold;
2. the finite truncations have an explicit tail estimate;
3. the magnetic phase dependence has a standard gauge-holonomy interpretation;
4. the Mertens quantity appears in a simple constant-mode expectation; and
5. the universal projection correction makes clear why the exact identity alone adds no analytic leverage.

Farey-map transfer operators are composition operators on function spaces, not this weighted graph Laplacian. The unweighted Farey graph also appears in network science, but its infinite-degree form fails under the construction studied here. The canonical exponent $\alpha=2$ also appears in related Ford-circle and Farey statistical-mechanics contexts. Similarity of exponents is motivation for further comparison, not proof of a shared operator mechanism.

Redheffer-type constructions provide another useful contrast: they place $M(N)$ in the determinant or norm behavior of explicit divisibility matrices. Here $M(N)$ enters through a position-weighted expectation of a constant kernel vector. The carriers and observables are different even though the same classical arithmetic function appears.

11. Reproducibility and Formal Boundaries

The accompanying deterministic checker constructs finite Farey graphs, verifies Hermiticity and positivity, checks the constant zero mode, evaluates the Farey-Mertens sum independently, verifies [17] numerically, replaces the positive spectrum while holding the kernel fixed, checks the finite trace-norm inequality, and evaluates a Rayleigh upper bound.

The current frozen run reports:

- 47 vertices and 91 edges at level 12;
- zero-mode error below $1.5\text{e-}17$;
- projection-identity vector error near $1.0\text{e-}13$;
- positive-spectrum replacement difference near $1.2\text{e-}13$; and
- all 12 governed checks and all 9 regression tests passing.

These computations are regression and reproducibility checks. The proofs of the finite identities are the algebra in the manuscript. The checker does not prove the infinite boundedness theorem, strong convergence, asymptotic gap bounds, the Mertens criterion, HSIT, SIT / Riemann, or RH.

Earlier versions named numerical scripts for gap fits, localization percentages, shell moments, and identity tables that were not present in the canonical repository audit. Those measurements are not used in this successor. They may be reintroduced only through a separately archived executable environment, exact inputs, commands, outputs, and a new evidence audit.

12. Open Problems

1. **Remove or justify the logarithm.** Determine whether $\lambda_2(L_N)$ is bounded below by cN^{-4} , or whether the logarithmic loss in [29] is asymptotically necessary.
2. **Find a noncircular arithmetic estimate.** Equation [20] is exact but universal. A useful advance would control the positional projection coefficient through additional Farey geometry without simply restating a bound on $M(N)$.
3. **Understand bottom-state geometry.** Determine whether low eigenvectors obey a stable continued-fraction-depth or cusp-localization law that can be proved independently of the removed numerical observations.
4. **Classify magnetic phases.** Identify whether natural modular cocycles produce limiting bottom-spectrum or holonomy statistics distinct from random phase fields.
5. **Compare with transfer operators.** Establish or rule out an explicit intertwining, limit, or coarse-graining relation between the weighted Laplacian and established Farey / Gauss transfer operators.
6. **Formalize the proved finite layer.** A proof assistant can certify the finite graph definitions, projection identity, canonical-path inequality, and finite arithmetic sum without implying the infinite or RH statements.

13. Claim and Version Ledger

This manuscript is a lossless scientific successor to Zenodo record 21231873, not a silent replacement of its historical text.

Claim	Final disposition
Slow-weight no-go for $\alpha \leq 1$	Proved on the stated form domain
Bounded positive operator for $\alpha > 1$	Proved
Strong finite-edge approximation	Proved for padded denominator truncations
Gauge covariance and closed-walk phase structure	Proved
Farey-Mertens exponential sum	Classical identity, directly derived
Commutator-Pythagoras formula	Preserved after correction to the universal projection theorem
Positive-spectrum mechanism for the identity	Retracted; positive spectral values cancel
Uniform canonical trace-class commutator	Proved
$O(N^{-4})$ gap upper bound	Proved
$c/(N^4 \log N)$ gap lower bound	Proved here with unspecified absolute $c > 0$
Fixed global traces as canonical Mertens carriers	Excluded for the stated function class

Claim	Final disposition
Earlier unarchived numerical measurements	Removed from the evidentiary chain
Physical SIT derivation, Hilbert-Polya operator, zeta-zero spectrum, SIT / Riemann bridge, or RH proof	Not established

The successor preserves the original questions, operator family, finite arithmetic identity, gauge structure, trace-class estimate, gap program, related-work context, reproducibility aim, and open-problem architecture. It changes the interpretation where exact algebra requires a change and promotes the lower-gap result only by reproducing its proof inside this paper.

14. Conclusion

The Farey graph does support a rigorous weighted Laplacian, but only after the infinite-degree problem is solved by sufficient denominator decay. The threshold $\alpha > 1$, the bounded operator, finite approximation, gauge-holonomy structure, trace-class commutator, and near-sharp N^{-4} gap scale are substantive results.

The arithmetic identity is equally exact and more limited than its earlier framing suggested. The constant mode and position unitary recover the Mertens sum, while the pseudoinverse commutator collapses to a universal kernel projection. This makes the RH equivalence transparent: it is a faithful operator-language restatement of the Mertens criterion, not a new proof of it.

That separation strengthens the paper. It leaves a defined operator rather than a slogan, explicit theorems rather than unsupported numerical claims, a reproduced gap proof rather than an unresolved internal reference, and open problems whose failure conditions are visible.

References

1. M. Blumberg, *The Weighted Farey Laplacian and the HSIT-Riemann Program*, Zenodo, 2026. [doi:10.5281/zenodo.21231873](https://doi.org/10.5281/zenodo.21231873)
2. G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, 6th ed., Oxford University Press, 2008. See Chapter XVI for Ramanujan sums and the standard Farey-sequence results.
3. E. C. Titchmarsh, revised by D. R. Heath-Brown, *The Theory of the Riemann Zeta-Function*, 2nd ed., Oxford University Press, 1986. See Section 14 for the classical Mertens criterion.
4. S. Kanemitsu and M. Yoshimoto, "Farey series and the Riemann hypothesis," *Acta Arithmetica* 75, 351-374, 1996. [doi:10.4064/aa-75-4-351-374](https://doi.org/10.4064/aa-75-4-351-374)
5. P. Diaconis and D. Stroock, "Geometric bounds for eigenvalues of Markov chains," *Annals of Applied Probability* 1, 36-61, 1991. [doi:10.1214/aoap/1177005980](https://doi.org/10.1214/aoap/1177005980)
6. L. Báez-Duarte, "A strengthening of the Nyman-Beurling criterion for the Riemann hypothesis," *Rendiconti Lincei - Matematica e Applicazioni* 14, 5-11, 2003. [arXiv:math/0202141](https://arxiv.org/abs/math/0202141)
7. J.-F. Burnol, "A lower bound in an approximation problem involving the zeros of the Riemann zeta function," *Advances in Mathematics* 170, 56-70, 2002. [arXiv:math/0103058](https://arxiv.org/abs/math/0103058)
8. X.-J. Li, "The positivity of a sequence of numbers and the Riemann hypothesis," *Journal of Number Theory* 65, 325-333, 1997.
9. E. Bombieri and J. C. Lagarias, "Complements to Li's criterion for the Riemann hypothesis," *Journal of Number Theory* 77, 274-287, 1999.
10. A. Connes, "Trace formula in noncommutative geometry and the zeros of the Riemann zeta function," *Selecta Mathematica* 5, 29-106, 1999.

11. D. H. Mayer, "The thermodynamic formalism approach to Selberg's zeta function for $\mathrm{PSL}(2, \mathbb{Z})$," *Bulletin of the American Mathematical Society* 25, 55-60, 1991.
12. J. Lewis and D. Zagier, "Period functions for Maass wave forms. I," *Annals of Mathematics* 153, 191-258, 2001.
13. C. Bonanno and S. Isola, "Spectral analysis of transfer operators associated to Farey fractions," 2007. arXiv:0708.0686
14. L. R. Ford, "Fractions," *American Mathematical Monthly* 45, 586-601, 1938.
15. B. Springborn, "The hyperbolic geometry of Markov's theorem on Diophantine approximation and quadratic forms," 2017. arXiv:1702.05061
16. M. Reed and B. Simon, *Methods of Modern Mathematical Physics I: Functional Analysis and II: Fourier Analysis, Self-Adjointness*, Academic Press.
17. T. Sunada, "A discrete analogue of periodic magnetic Schrödinger operators," *Contemporary Mathematics* 173, 283-299, 1994.
18. Z. Zhang and F. Comellas, "Farey graphs as models for complex networks," *Theoretical Computer Science* 412, 865-875, 2011.
19. A. E. Ingham, "On two conjectures in the theory of numbers," *American Journal of Mathematics* 64, 313-319, 1942.
20. J. Franel, "Les suites de Farey et le problème des nombres premiers," and E. Landau, "Bemerkungen zu der vorstehenden Abhandlung von Herrn Franel," *Nachrichten von der Gesellschaft der Wissenschaften zu Göttingen*, 1924.
21. R. M. Redheffer, "Eine explizit lösbare Optimierungsaufgabe," *Internationale Schriftenreihe zur Numerischen Mathematik* 36, 1977.